

Sri Lakshmi Bathina

 sbathina02@gmail.com |  (484) 582-9690 |  [LinkedIn](#) |  [GitHub](#)

PROFESSIONAL SUMMARY

Data Science graduate with hands-on experience building cloud-based data pipelines, preparing large datasets, and developing machine learning models for real-world use cases. Skilled in Python, SQL, AWS, and ETL automation. Strong foundation in statistical analysis, predictive modeling, and translating data into clear business insights. Eager to contribute to data engineering, analytics, or ML-focused roles

EDUCATION

University of Massachusetts

Master of Science in Data Science, GPA: 3.6/4.0

May, 2025

Dartmouth, MA

Gandhi Institution of Technology

Bachelor of Technology in Computer Science, GPA: 3.72/4.0

May, 2023

Visakhapatnam, India

AWS Certified AI Practitioner (In Progress)

PROJECTS

~ Representative projects developed during the course of my master's studies.

Exercise & Calorie Coach Orchestrator Agent (Gemini + ADK) – Kaggle Agents Intensive Capstone Project

Tools: Python, Pandas, Kaggle Notebooks, Google Gemini, Google Agent Development Kit (ADK)

- Built a Gemini + ADK “fitness coach” agent over a calories-burned exercise dataset that recommends personalized workout plans based on user weight, calorie target, and time budget.
- Implemented a Python function tool to compute optimal activities and durations (calories per minute, max calories in a time window) and exposed it as an ADK tool for automatic function calling.
- Added lightweight observability by logging each tool invocation and testing multiple user scenarios to verify sensible, goal-aligned workout recommendations.

AI-Powered Chest X-ray Diagnostic System

Tools: Python, PyTorch, TorchXRayVision, Grad-CAM, scikit-learn, pandas, NumPy, Streamlit

- Developed DenseNet121 model to classify 14 chest conditions, achieving ~87 % accuracy on NIH dataset of 100 K images.
- Enhanced interpretability via Grad-CAM heatmaps and LLM-based summaries, improving stakeholder clarity by 40 %.
- Deployed interactive Streamlit app for real-time inference and explainability, reducing manual analysis time by ~50 %.

Parallel vs Non-Parallel Random Forest Model Evaluation

Tools: scikit-learn, multiprocessing, GridSearchCV

- Implemented both parallel and non-parallel Random Forest pipelines to evaluate scalability and runtime trade-offs across CPU cores.
- Optimized model training using multiprocessing (`n_jobs`), reducing overall runtime by 60% while maintaining prediction stability.
- Conducted hyperparameter tuning with GridSearchCV, improving the F1-score by 8% and ensuring reproducible model benchmarking results.

Spotify Most Streamed Songs Analysis

Tools: Python, Pandas, Seaborn, Matplotlib, PowerBI

- Cleaned and preprocessed 50,000+ tracks to perform exploratory data analysis (EDA) on listener patterns, popularity metrics, and audio features.
- Identified key correlations between popularity, acousticness, and energy, visualizing temporal trends in song duration and release frequency.
- Designed an interactive Power BI and Matplotlib dashboard summarizing genre performance, artist dominance, and trend forecasts for strategic insights.

INTERNSHIP EXPERIENCE

Data Analytics Intern – Phoenix Global

Remote | Mar 2022 – Sep 2022

Tools: Python, SQL, Power BI, Excel, Jupyter, Git, Agile

- Engineered Python + SQL ETL pipelines processing 10 K + user records, automating ingestion and transformation to reduce reporting latency by 30 %.
- Designed Power BI dashboards and SQL semantic models that cut manual reporting effort by 10 + hours weekly and improved data accuracy across teams.
- Partnered in Agile sprints to gather requirements and validate data flows, boosting pipeline reliability by ~25 % through iterative testing and QA.

SKILLS

- **Languages & ML Tools:** Python (NumPy, pandas, scikit-learn, XGBoost, PyTorch), R
- **Data Engineering & Cloud:** AWS (S3, EC2, Glue, Lambda), PySpark, GCP (Gemini, ADK), Azure, Databricks
- **Visualization & Analytics:** Tableau, Power BI, Fabric, Excel (Power Query), Matplotlib, Seaborn
- **Databases & Workflow:** SQL, MySQL, Git, Jupyter, Kaggle Notebook, VS Code